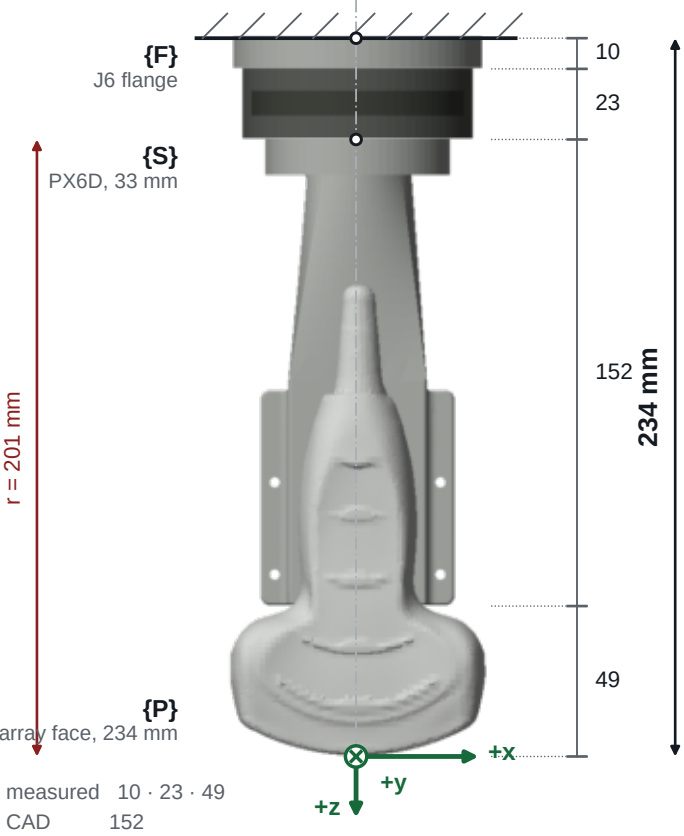


(a) Tool stack, seen in the image plane

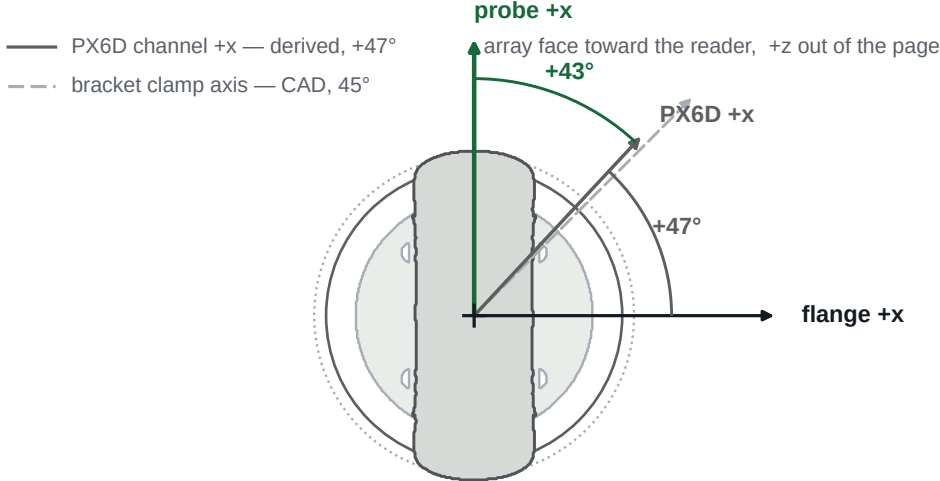
flange → adapter → PX6D → bracket → probe, all coaxial

image plane in the page; elevational into it



(b) Axial registration

who is turned by how much about the common +z



$${}^F R_S = R_z(+90^\circ) \cdot R_z(-43^\circ) = R_z(+47^\circ)$$

CAD clamp 45° vs. as-built 43°: the 2° is assembly, not design

(c) Probe frame {P} and the image plane Π

DESIGN_NOTES §4.1 — +z into tissue, +x along the array

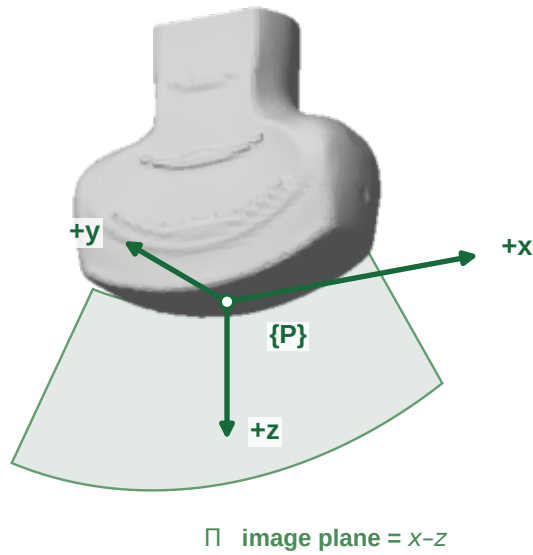
{P} origin = array face centre, 234 mm from J6

convex array R = 82.6 mm, sector ±29° (from the solid)

[!] Fr5Model.tsx draws this triad in the flange frame — the +90° is missing from it

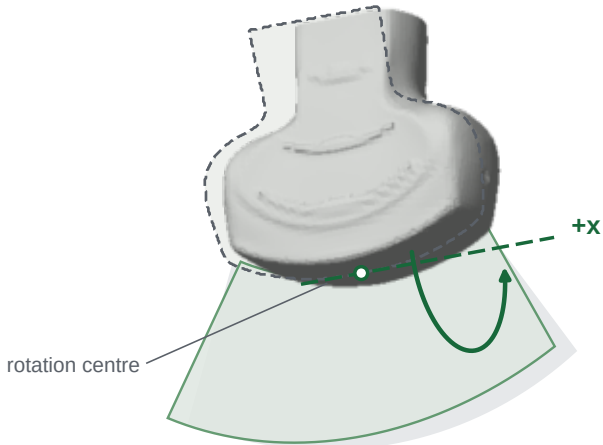
console triad — told apart by length, not colour

+x	lateral, inside the image plane	75 mm
+y	elevational, out of the plane	55 mm
+z	axial, into the tissue	38 mm



(d) roll ω_x

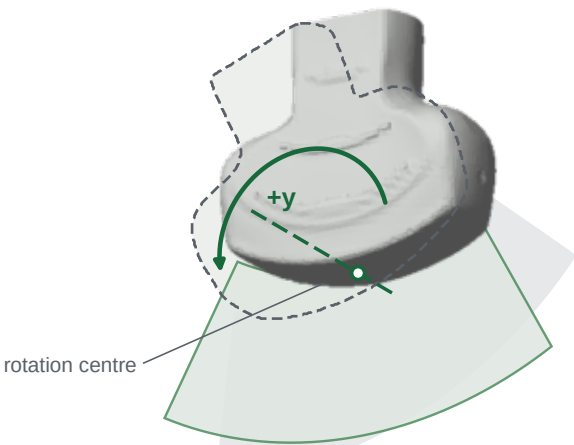
about the lateral axis → out-of-plane tilt (fanning)



new section · admittance drives $M_x \rightarrow 0$

(e) pitch ω_y

about the elevational axis → in-plane rock (heel-toe)

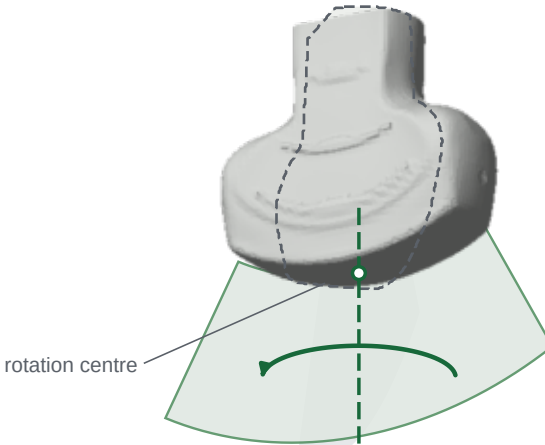


drawn at 22°

same section · admittance drives $M_y \rightarrow 0$

(f) yaw ω_z

about the beam axis → image-plane rotation



drawn at 22°

new section · image policy holds this axis

drawn at 45°

Rotation centre is the tool-frame origin (§4.5). An offset δ between it and the true array face turns a commanded ω into contact slip $v_{slip} = \omega \times \delta$: $\omega = 0.2$ rad/s with $\delta = 20$ mm gives 4 mm/s, 40% of the 10 mm/s contact translation budget.

teleop mapping: $A = R_z(\text{tip_roll}) \cdot \text{diag}(+1, -1, -1)$, $\text{tip_roll} = 0^\circ$